Syimyk (Sam) Khamdamov

Boston, US (Open to Relocation) | syimykkhamdamov@gmail.com | +1 774 722 7441

semthedev | Syimyk Khamdamov | sk_khamdamov | Syimyk Khamdamov

Summary

Experience

AI Researcher — *January 2026 — May 2026*

International Laboratory for Intelligent Systems and Structural Analysis (ISSA), Higher School of Economics (HSE), Faculty of Computer Science — Moscow, Russia

Advisor: Joseph Seungmin Jin | In collaboration with Huawei

- Designed an interpretability pipeline using Python, PyTorch, Formal Concept Analysis (FCA) and Caspailleur to project GloVe 300D embeddings into a structured 3D space; mapped these representations into Hasse diagrams, reducing dimensionality by 99% and transforming black-box embeddings into clear, interconnected networks of word and sentence meanings.
- Benchmarked linear-complexity Mirkin Attention within the Noether-Symmetric Transformer (NST) framework using PyTorch, evaluating it against standard quadratic self-attention and reducing theoretical computational complexity from $O(N^2)$ to $O(N)$ for long-sequence modeling tasks.

MLOps Engineer Intern — *September 2025 — December 2025*

Procter & Gamble — Moscow, Russia

- Engineered a centralized feature store (1.5M+ records) for Eastern Europe & Central Asia, powering forecasting and ML models used by 280+ cross-functional stakeholders.
- Built and maintained daily ETL/ELT pipelines using PySpark and Databricks with Delta Lake storage: implemented Bronze/Silver layers, SCD Type 2 history, and integrated automated feature engineering and text embedding generation (Hugging Face Transformers) into the data flow.
- Engineered a comprehensive Data Quality framework (70+ checks) combining robust statistical profiling, clustering, and deep learning anomaly detection to isolate outliers, assessing their direct financial impact on product sales.
- Managed end-to-end deployment across DEV, TEST, and PROD environments using strict Git workflows, CI/CD pipelines, and unit tests; data centralization reduced the onboarding time for new ML data products by 3 weeks.

Data Analyst / Data Scientist Intern — *December 2024 — February 2025*

Korus Consulting — Moscow, Russia

- Reduced ingredient overfill by 86% against baseline by building and comparing ML models (XGBoost, CatBoost, LightGBM, scikit-learn) to predict dosing deviations and selecting the best-performing one for production use.
- Automated data cleaning pipelines that eliminated 97% of anomalous readings by formulating and testing hypotheses (statistical methods) on sensor data and implementing rule-based resolution logic in Python.
- Built interactive dashboards in Apache Superset and Power BI to visualize dosing deviations and analysis results, replacing 4 recurring manual reports and saving 12 hours/week previously spent on ad-hoc requests to the IT team.

Technical Skills

- **Programming languages:** Python, SQL, C++, C#, JavaScript, Bash
- **Databases & warehouses:** PostgreSQL, MySQL, ClickHouse, Greenplum, Oracle, MongoDB
- **Data engineering & cloud:** Apache Airflow, Apache Spark, Apache Hadoop, Databricks, Azure Data Factory, Docker, Kubernetes, Git
- **BI & visualization:** Power BI, Tableau, Apache Superset, Plotly, Matplotlib, Seaborn, Excel
- **ML & DS libraries:** pandas, NumPy, scikit-learn, statsmodels, SciPy, XGBoost, CatBoost, LightGBM, PyTorch, TensorFlow, Hugging Face Transformers, Sentence-Transformers, LangChain, LlamaIndex, OpenCV, FastAPI
- **ML & DS tools:** MLflow, Optuna, Qdrant, Jupyter Notebook
- **ML & Data Science areas:** Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), Large Language Models (LLM), Computer Vision (CV), Retrieval-Augmented Generation (RAG), Information Retrieval (IR), Recommender Systems, Time Series Analysis, A/B Testing, Statistical Hypothesis Testing, Feature Engineering, Word Embeddings, Model Interpretability, Formal Concept Analysis (FCA)
- **Methodologies:** Agile/Scrum, CI/CD, Unit testing, Experimentation
- **Languages:** Native — Kyrgyz, Russian; Fluent — Turkish, Kazakh, Uzbek; Professional — English; Basic — Spanish, Italian

Hackathons & Competitions

1st place out of 30 teams — "AI Champ" ML Hackathon (Central University & ITMO), Okko Video RAG Challenge — *March 2026*

Team FrameTamers (4 members) — certificate, code

- Built an end-to-end Russian-language Video RAG pipeline that retrieves relevant movie fragments from natural-language queries (final score 0.57738): GigaAM-v3 for Russian ASR, ensemble of fine-tuned BGE-M3 and Qwen3-Embedding over 9,476 video chunks with cosine similarity search, Dynamic HyDE and Answer Augmentation for query expansion, deployed as a FastAPI and Streamlit web app with video file streaming. *[Python, PyTorch, BGE-M3, Qwen3-Embedding, GigaAM-v3, HyDE, FastAPI, Streamlit, RAG, IR, Docker, Kubernetes]*

Publications

Development of an Automated Information System for Scheduling in Technical Universities Based on a Genetic Algorithm *November 2024*

- III All-Russian Scientific Conference "Artificial Intelligence in Automated Control and Data Processing Systems"

IIASU'24 — "Artificial Intelligence in Automated Control and Data Processing Systems". Proceedings of the III All-Russian Scientific Conference (Moscow, October 30 — November 1, 2024), vol. 2, pp. 379–387

Projects

Computer Vision: Classification and Detection

- Image classification: trained models for MNIST (MLP, PyTorch) and Fashion-MNIST (Dense/Dropout, Keras); built a full preprocessing pipeline (normalization, splitting, DataLoader/generators), configured training (Adam/SGD), validation, and metric logging.
- Object recognition on CIFAR-10: designed and trained a CNN (Conv2D + MaxPooling, Dropout, Flatten, Dense); implemented one-hot label encoding and automated evaluation on the test set.
- Face detection: integrated MediaPipe Tasks (BlazeFace, `.tflite`) and OpenCV; implemented conversion from normalized coordinates to pixels and drawing bounding boxes and keypoints.
- Image processing: built utilities for *resize/rotate/translate*, color space conversion, and before/after visualization; explored spatial filtering (median and Laplacian filters) for denoising and sharpening.
- *Stack:* Python, PyTorch, TensorFlow/Keras, OpenCV, MediaPipe, NumPy, Matplotlib
- Code

Impact of Activity and Sleep on Weight Change (Exploratory and Predictive Analysis)

- Studied how physical activity level and sleep quality affect participants' weight change; performed data cleaning and merging, distribution visualization, correlation and clustering analysis.
- Built a regression model to predict participants' final weight; identified the most significant factors (current weight and weight change) influencing the final outcome.
- Implemented multiple visualization methods: heatmaps, boxplots, bar plots, line plots; applied KMeans clustering and interpreted cluster behavior based on stress level, program duration, and calorie surplus/deficit.
- *Stack:* Python (pandas, seaborn, matplotlib, scikit-learn), Jupyter Notebook
- Code

Sales Analytics Pipeline (Greenplum, Airflow, Superset, ClickHouse)

- Designed an end-to-end ETL pipeline to ingest data from CSV and PostgreSQL into Greenplum using gpfdist and PXF; configured external tables, distribution keys, partitioning, and sales data marts.
- Implemented partition management functions (upsert/load) and developed Apache Airflow DAGs for sequential temporary table processing and aggregate loading into ClickHouse.
- Configured replication and table partitioning in ClickHouse, integrating it with Apache Superset as a high-performance serving layer for analytical dashboards.

Education

Politecnico di Milano, Master of Science in Computer Science and Engineering — Milan, Italy *(QS World Rank: #87, Global Top-20 in Engineering & Technology, #1 in Italy)* *September 2026 — present (expected graduation: August 2028)*
Computer Science and Engineering program, Data Science and Artificial Intelligence tracks (English-taught)

Higher School of Economics, Faculty of Computer Science partnered with Yandex — Moscow, Russia *(QS World Top-400, #1 in Russia per Forbes 2025 employer reputation ranking)* *September 2025 — present (expected graduation: August 2027)*
Master's program in "Applied Mathematics and Informatics" jointly with Yandex School of Data Analysis (YSDA) Data Science track (English-taught)

Russian Biotechnological University (ROSBIOTECH) — Moscow, Russia *September 2021 — August 2025*
"Bachelor's degree in "Informatics and Computer Engineering" - Graduated with Honors Specialization: Artificial Intelligence for Control of Technological Complexes